

E-Commerce & Retail Sales Data Analytics (DA) Report

Comprehensive End-to-End Analysis Across Sales Performance, Customer Insights, and Product & Discount Strategy

1. Problem Statement

The retail business seeks to analyze its historical sales and customer transaction data to evaluate overall operational performance, identify profitability drivers, assess geographical and channel trends, and support data-driven strategic decision-making. The primary objective is to maximize revenue and profit margins by optimizing product categories, sales channels, and geographical distribution while effectively managing promotional discounts and operational costs across all business segments.

2. Dataset Overview

The dataset comprises 5,000 transactional order records capturing granular sales, geography, customer satisfaction, and financial metrics. Below is the detailed breakdown of key fields:

- **Order Details (Order_ID, Order_Date):** Unique identifier for each customer purchase order.
- **Customer Details (Customer_Name, Customer_Rating):** Customer identification and rating scores (1.0 to 5.0).
- **Geographic Details (City, Country):** Store location details spanning global cities and countries.
- **Product Details (Product_Category, Product_Name, Quantity, Unit_Price):** Product classification across Beauty, Electronics, Fashion, Home & Living, and Sports.
- **Sales & Cost Metrics (Discount_Percent, Revenue, Cost, Profit):** Financial figures measuring discount percentages, sales revenue, total product cost, and net profit.
- **Channel & Payment (Sales_Channel, Payment_Method):** Fulfillment channel (Online, Retail Store, Mobile App) and payment gateway.

3. Data Preprocessing Steps

Prior to building the interactive dashboards and establishing DAX measures, the raw transactional dataset underwent rigorous preprocessing steps:

- Verified order records and removed duplicate entries to ensure exact transaction counts.
- Audited all categorical and numeric fields for missing values; no null values were found.
- Standardized Order_Date into a unified datetime format and generated Date hierarchies (Year, Quarter, Month).

- Derived Total Cost per line item ($\text{Cost} = \text{Revenue} - \text{Profit}$) and calculated total promotional discount values.
- Validated data types for Revenue, Cost, Profit, Discount_Percent, Quantity, and Customer_Rating to prevent aggregation errors.
- Standardized column naming conventions across all data models for consistent DAX measure authoring.

4. Sheets / Pages Created in the Report

4.1 Executive Overview Dashboard (Sales Performance Dashboard)

Purpose: Provides an executive-level summary of overall business performance, revenue trends, and primary distribution channels.

- Total Revenue: ₹ 24,981,982.37
- Total Profit: ₹ 7,144,699.47
- Total Orders: 5,000
- Average Customer Rating: 3.0 / 5.0
- Total Products Sold: 22,386 units
- Monthly Revenue Trend (Line Chart showing revenue fluctuations across months)
- Revenue by Product Category & Sales Channel (Horizontal bar & donut breakdown)
- Geographic Revenue Distribution Map

4.2 Customer & Sales Insights Dashboard

Purpose: Delivers granular insights into customer demographics, purchasing behavior, order profitability, and payment preferences.

- Total Unique Customers: 4,843
- Average Order Value (AOV): ₹ 4,996.40
- Overall Profit Margin: 28.60%
- Revenue vs. Profit Trend Comparison
- Category Profit Contribution Share (%)
- Payment Method Breakdown (Credit Card, Debit Card, Bank Transfer, PayPal, Cash on Delivery)
- Top 10 Products and Top 10 Customers by Revenue
- Scatter plot showing Customer Rating vs. Order Profitability correlation

4.3 Product & Discount Strategy Dashboard

Purpose: Evaluates pricing effectiveness, cost structure, product category volumes, and the net impact of discounts on bottom-line profit.

- Total Product Cost: ₹ 17,837,282.90
- Average Discount Percentage: 12.56%

- Total Discount Value Granted: ₹ 3.60M
- Average Profit Per Order: ₹ 1,428.94
- Average Quantity Per Order: 4.48 units
- Cost vs. Revenue Comparison by Product Category
- Sales Channel Discount Share (Treemap distribution)
- Top 10 Cities by Revenue & Top 10 Most Discounted Products

5. DAX Measures Used

The analytical model utilizes standard DAX (Data Analysis Expressions) measures to calculate metrics dynamically across slicers and page filters:

5.1 Basic Measures

```
Total Revenue = SUM(Sales_Dataset[Revenue])
Total Profit = SUM(Sales_Dataset[Profit])
Total Quantity = SUM(Sales_Dataset[Quantity])
Total Cost = SUM(Sales_Dataset[Cost])
```

5.2 Derived Measures

```
Total Orders = DISTINCTCOUNT(Sales_Dataset[Order_ID])
Total Customers = DISTINCTCOUNT(Sales_Dataset[Customer_Name])
Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)
Profit Margin = DIVIDE([Total Profit], [Total Revenue], 0)
Average Discount Percent = AVERAGE(Sales_Dataset[Discount_Percent])
Average Customer Rating = AVERAGE(Sales_Dataset[Customer_Rating])
```

5.3 Time Intelligence Measures

```
Revenue YoY Growth =
CALCULATE(
    [Total Revenue],
    SAMEPERIODLASTYEAR(Sales_Dataset[Order_Date])
)
```

5.4 Ranking Measures

```
Rank Products by Revenue =
RANKX(
    ALL(Sales_Dataset[Product_Name]),
    [Total Revenue],
    ,
    DESC
)
```

6. Key Insights & Findings

- **Category Performance:** Home & Living generates the highest overall revenue (₹ 5,351,720.47) and net profit (₹ 1,521,121.36), followed closely by Electronics (₹ 4,984,890.43).
- **Sales Channel Distribution:** Sales are almost equally distributed across Online (34.0%), Retail Store (33.8%), and Mobile App (32.2%), demonstrating balanced omni-channel customer adoption.
- **Geographic Dominance:** The USA (₹ 3.96M Revenue) and India (₹ 3.83M Revenue) lead global sales, contributing to nearly 31.2% of overall company revenue.
- **Profit Margin Stability:** Profit margins remain remarkably consistent across all five categories at 28.0% to 29.1%, proving stable pricing power.
- **Discount Impact:** An average discount rate of 12.56% maintains a healthy balance between driving purchase volume and preserving healthy profit margins.

7. Final Assumptions & Business Recommendations

Assumptions

- Recorded monetary figures represent accurate transactional data in Indian Rupees (₹).
- General overhead, return penalties, and logistics costs are excluded from net profit calculations.
- Recorded customer ratings accurately capture post-purchase customer satisfaction.

Recommendations

- **Strategic Marketing:** Allocate additional promotional budget to high-yield geographic markets (USA, India, Pakistan) and leading product lines (Home & Living, Electronics).
- **Discount Optimization:** Cap maximum promotional discounts at 15% for lower-margin sub-items to protect the target 28.60% overall profit margin.
- **Channel Engagement:** Implement exclusive mobile-app user loyalty incentives to boost Mobile App revenue share closer to Online and Retail Store levels.
- **Customer Experience:** Investigate order fulfillment and product quality issues for orders with low rating scores (1–2 stars) to improve customer retention.

8. Final Conclusion (DA Report Summary)

The Superstore Data Analytics Report provides executive stakeholders with actionable visibility into revenue drivers, product margins, geographic dynamics, and promotional efficiency. By leveraging interactive dashboards alongside structured DAX calculations, business leaders can quickly isolate performance bottlenecks and capitalization opportunities. Implementing the recommended operational and marketing strategies will significantly enhance long-term profitability and market positioning.